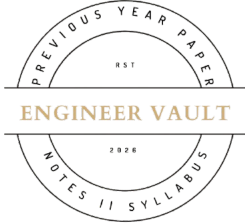


Total No. of Printed Pages: 02

**SUBJECT CODE NO: - RNP-8015****FACULTY OF SCIENCE AND TECHNOLOGY****F.E. (Group B) CSE / AI & ML / EEE (NEP) Sem - II****Examination June / July 2025****Programming for Problem Solving****[Time: 2.00 Hours]****[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

1) Q.no. 1 is compulsory.

2) Solve any TWO questions from Q.no. 2, 3, 4 and 5

SECTION A**Q1 Solve any Five****10**

A. What will be the value of x in the following Python expression?

 $x = \text{int}(43.55 + 2/2)$

- a) 43 b) 44 c) 22 d) 23

B. Which of the following is true for variable names in Python?

- a) underscore and ampersand are the only two special characters allowed
 b) unlimited length
 c) all private members must have leading and trailing underscores
 d) none of the mentioned

C. What is the purpose of the range () function in Python?

- a) To generate a sequence of numbers
 b) To iterate over items in a sequence
 c) To define a function
 d) To execute a block of code repeatedly until a condition is false

D. What does *list1* + *list2* do?

- a) Modifies *list1* b) Merges *list1* and *list2*
 c) Appends *list2* to *list1* d) Error

E. Which type of Programming does Python support?

- a) object-oriented programming
 b) structured programming
 c) functional programming
 d) all of the mentioned

F. Which method removes whitespace from the beginning and end of a string?

- a) strip() b) remove() c) trim() d) delete()

G. Which operation is used to find common elements in sets?

- a) intersection () b) union ()
 c) difference () d) symmetric_difference()

SECTION B

- Q2** A. Explain the applications of Python programming language. **05**
 B. Explain bitwise operators in python. **05**
- Q3** A. With the help of syntax explain how to use elif statement in python also write program to display day of the week using elif. **05**
 B. Explain while loop in python. Also write a program to display first 10 Fibonacci numbers using while loop **05**
- Q4** A. Explain recursive function. Write a code to display factorial of a number using recursive function **05**
 B. Explain role of + and * in string manipulation. **05**
- Q5** A. Differentiate between set and list in python **05**
 B. Write a Python program to update and delete elements in a dictionary. **05**

